

◆ SECTION 3: Density & Unit Cell Calculations (Q41–70)

41. Density of a solid is given by:
- A) $\rho = \frac{ZM}{a^3 N_A}$
 - B) $\rho = \frac{Ma^3}{ZN_A}$
 - C) $\rho = \frac{a^3 N_A}{ZM}$
 - D) $\rho = \frac{ZM}{Ma^3}$
42. Formula for number of atoms per unit cell (Z) in FCC:
- A) 1
 - B) 2
 - C) 4
 - D) 8
43. Formula for number of atoms per unit cell (Z) in BCC:
- A) 1
 - B) 2
 - C) 4
 - D) 8
44. Formula for number of atoms per unit cell (Z) in SC:
- A) 1
 - B) 2
 - C) 4
 - D) 8
45. Molar mass of copper is 63.5 g/mol, radius = 0.128 nm, FCC structure. Calculate density. (MCQ options approximate)
- A) 8.92 g/cm³
 - B) 7.87 g/cm³
 - C) 9.2 g/cm³
 - D) 10 g/cm³
46. Number of formula units in NaCl unit cell:
- A) 2
 - B) 4
 - C) 6
 - D) 8
47. Edge length of SC unit cell is 4 Å. Atomic mass = 20 g/mol, calculate density.
- A) 2 g/cm³
 - B) 4 g/cm³
 - C) 5 g/cm³
 - D) 6 g/cm³
48. Atomic packing factor (APF) =
- A) Volume of atoms / Volume of unit cell
 - B) Volume of unit cell / Volume of atoms
 - C) Mass / Volume
 - D) Number of atoms × radius
49. In FCC, relation between atomic radius r and edge length a:
- A) $a = 2r$
 - B) $a = 2\sqrt{2} r$

- C) $a = 4r$
D) $a = r$
50. Density of BCC metal is generally:
A) Higher than SC
B) Lower than SC
C) Same as SC
D) Zero
51. NaCl has FCC lattice. Number of Na^+ ions per unit cell:
A) 1
B) 2
C) 4
D) 6
52. Number of Cl^- ions in NaCl unit cell:
A) 1
B) 2
C) 4
D) 6
53. If edge length a of cubic unit cell doubles, density:
A) Doubles
B) Halves
C) Reduces to one-fourth
D) Remains same
54. Mass of atoms per unit cell =
A) $Z \times \text{atomic mass} / N_A$
B) $N_A / (Z \times \text{atomic mass})$
C) $Z \times N_A / \text{atomic mass}$
D) $\text{atomic mass} / Z$
55. Formula to calculate lattice constant from density:
A) $a = (Z M / \rho N_A)^{1/3}$
B) $a = (\rho N_A / Z M)^{1/3}$
C) $a = (\rho N_A / Z M)^{1/2}$
D) $a = (Z M / \rho N_A)$
56. Relation between radius r and octahedral void radius R_o :
A) $R_o = 0.414 r$
B) $R_o = 0.225 r$
C) $R_o = 0.155 r$
D) $R_o = 0.732 r$
57. Relation between radius r and tetrahedral void radius R_t :
A) $R_t = 0.414 r$
B) $R_t = 0.225 r$
C) $R_t = 0.155 r$
D) $R_t = 0.732 r$
58. Tetrahedral voids per atom in FCC:
A) 1
B) 2
C) 4
D) 6

59. Octahedral voids per atom in FCC:
- A) 1
 - B) 2
 - C) 3
 - D) 4
60. Packing efficiency is directly proportional to:
- A) Atomic mass
 - B) Atomic radius
 - C) Volume occupied by atoms
 - D) Edge length
61. Density of a unit cell increases when:
- A) Atomic mass decreases
 - B) Number of atoms increases
 - C) Volume of unit cell increases
 - D) Both A and C
62. HCP structure: density formula uses:
- A) 6 atoms per unit cell
 - B) 8 atoms per unit cell
 - C) 4 atoms per unit cell
 - D) 2 atoms per unit cell
63. Ionic solids density depends on:
- A) Atomic mass
 - B) Number of ions per unit cell
 - C) Edge length
 - D) All of the above
64. SC structure has lowest density due to:
- A) Large voids
 - B) Small number of atoms
 - C) Low packing efficiency
 - D) All of the above
65. Radius of ion in octahedral void depends on:
- A) Atomic radius of host atom
 - B) Coordination number
 - C) Both A & B
 - D) None
66. Tetrahedral void size is:
- A) Larger than octahedral
 - B) Smaller than octahedral
 - C) Equal to octahedral
 - D) Zero
67. Which lattice has highest packing efficiency?
- A) SC
 - B) BCC
 - C) FCC
 - D) None
68. Calculate approximate radius of atom if edge length of FCC metal = $4.05 \text{ \AA}$:
- A) $1.43 \text{ \AA}$
 - B) $2 \text{ \AA}$

- C) 1 Å
 - D) 1.5 Å
69. Number of formula units in CsCl unit cell:
- A) 1
 - B) 2
 - C) 4
 - D) 8
70. CsCl structure is:
- A) FCC
 - B) BCC
 - C) Simple cubic
 - D) HCP

◆ SECTION 4: Imperfections in Solids (Q71–90)

71. Schottky defect is:
- A) Vacancy defect
 - B) Interstitial defect
 - C) Substitutional defect
 - D) Frenkel defect
72. Frenkel defect is:
- A) Vacancy defect
 - B) Interstitial defect
 - C) Both A & B
 - D) None
73. Colour of NaCl changes due to:
- A) Schottky defect
 - B) Frenkel defect
 - C) F-centres
 - D) None
74. Effect of Schottky defect on density:
- A) Increases
 - B) Decreases
 - C) No effect
 - D) Zero
75. Effect of Frenkel defect on density:
- A) Increases
 - B) Decreases
 - C) No effect
 - D) Zero
76. Edge dislocation affects:
- A) Electrical properties
 - B) Mechanical properties
 - C) Magnetic properties
 - D) Density
77. Screw dislocation is a:
- A) Point defect

- B) Line defect
 - C) Surface defect
 - D) None
78. Colour centres are formed due to:
- A) Missing anions
 - B) Extra cations
 - C) Both
 - D) None
79. Intrinsic defects are caused by:
- A) Impurities
 - B) Thermal agitation
 - C) Pressure
 - D) Magnetic field
80. Extrinsic defects are caused by:
- A) Thermal energy
 - B) Impurities
 - C) None
 - D) Pressure
81. Which defect is common in AgCl?
- A) Schottky
 - B) Frenkel
 - C) Both
 - D) None
82. Defects increase with:
- A) Temperature
 - B) Pressure
 - C) Magnetic field
 - D) Voltage
83. Grain boundary is:
- A) Line defect
 - B) Surface defect
 - C) Point defect
 - D) Bulk defect
84. Stacking fault is:
- A) Line defect
 - B) Surface defect
 - C) Point defect
 - D) Volume defect
85. Dislocations make metals:
- A) Harder
 - B) Softer
 - C) Brittle
 - D) None
86. Vacancy in ionic solids:
- A) Reduces density
 - B) Increases density
 - C) No effect
 - D) None

87. Defects affect:
- A) Mechanical properties
 - B) Electrical conductivity
 - C) Diffusion
 - D) All of the above
88. Frenkel defect is common in:
- A) NaCl
 - B) AgCl
 - C) KCl
 - D) CsCl
89. Schottky defect is common in:
- A) NaCl
 - B) ZnS
 - C) AgCl
 - D) CsCl
90. F-centres are responsible for:
- A) Colour
 - B) Hardness
 - C) Density
 - D) Magnetic property

◆ SECTION 5: Electrical & Magnetic Properties (Q91–120)

91. Conductivity of metals:
- A) High
 - B) Low
 - C) Zero
 - D) Medium
92. Diamond is:
- A) Conductor
 - B) Insulator
 - C) Semiconductor
 - D) Superconductor
93. Si is:
- A) Conductor
 - B) Semiconductor
 - C) Insulator
 - D) None
94. Intrinsic semiconductor conductivity increases with:
- A) Temperature
 - B) Pressure
 - C) Voltage
 - D) Magnetic field
95. N-type semiconductor has:
- A) Extra electrons
 - B) Extra holes

- C) Neutral
 - D) None
96. P-type semiconductor has:
- A) Extra electrons
 - B) Extra holes
 - C) Neutral
 - D) None
97. Diamagnetic materials are:
- A) Weakly repelled by magnetic field
 - B) Strongly attracted
 - C) Neutral
 - D) Both
98. Paramagnetic materials:
- A) Weakly attracted
 - B) Weakly repelled
 - C) Strongly attracted
 - D) None
99. Ferromagnetic materials:
- A) Strongly attracted
 - B) Weakly attracted
 - C) Neutral
 - D) Repelled
100. Curie temperature is:
- A) Temperature at which ferromagnet becomes paramagnetic
 - B) Melting point
 - C) Boiling point
 - D) Freezing point
101. Superconductors have:
- A) Zero resistance
 - B) High resistance
 - C) Medium resistance
 - D) Infinite resistance
102. Meissner effect:
- A) Expulsion of magnetic field
 - B) Attraction to magnet
 - C) Electrical conductivity
 - D) Resistance
103. Hall effect is used to determine:
- A) Type of semiconductor
 - B) Magnetic field
 - C) Resistivity
 - D) Current
104. Piezoelectric solids generate:
- A) Voltage on stress
 - B) Heat
 - C) Magnetic field
 - D) None

105. Examples of piezoelectric solids:
- A) Quartz, Rochelle salt
 - B) NaCl
 - C) Diamond
 - D) Al
106. Magnetic domains are:
- A) Regions of aligned magnetic moments
 - B) Crystals
 - C) Defects
 - D) None
107. Paramagnetic susceptibility is:
- A) Positive
 - B) Negative
 - C) Zero
 - D) Infinite
108. Diamagnetic susceptibility is:
- A) Positive
 - B) Negative
 - C) Zero
 - D) Infinite
109. Conductivity of semiconductors increases with:
- A) Doping
 - B) Temperature
 - C) Both
 - D) None
110. Intrinsic semiconductors:
- A) Pure
 - B) Doped
 - C) Conductors
 - D) Insulators
111. Extrinsic semiconductors:
- A) Doped
 - B) Pure
 - C) Metals
 - D) Insulators
112. Example of ferromagnetic material:
- A) Fe
 - B) Cu
 - C) Na
 - D) Al
113. Example of paramagnetic material:
- A) Al
 - B) Fe
 - C) Ni
 - D) Co
114. Example of diamagnetic material:
- A) Cu
 - B) Fe

- C) Ni
 - D) Co
115. Conductivity of metals decreases with:
- A) Temperature increase
 - B) Temperature decrease
 - C) Pressure increase
 - D) None
116. Ferromagnetic materials retain magnetization:
- A) Yes
 - B) No
 - C) Sometimes
 - D) Never
117. Semiconductors are:
- A) Conductors
 - B) Insulators
 - C) Intermediate conductivity
 - D) Superconductors
118. Colour centres occur due to:
- A) Missing anions
 - B) Extra cations
 - C) Both
 - D) None
119. Superconductivity is destroyed above:
- A) Critical temperature
 - B) Melting point
 - C) Boiling point
 - D) Curie temperature
120. Magnetic hysteresis curve is for:
- A) Ferromagnetic materials
 - B) Paramagnetic materials
 - C) Diamagnetic materials
 - D) Superconductors